

Real-Time Fractional Tracking (R-TFT): Resonance Compression and Containment System (RCCS) – Refined Edition

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Abstract

This paper presents the refined, fully self-contained framework of the Resonance Compression and Containment System (RCCS), embedding golden-ratio coherence and resonance ethics containment at the mathematical and structural level. Designed as a definitive evolution of the original RCCS model, this version incorporates recursive filtering, ϕ -coherence validation, and ethical constraints to ensure only convergent, non-malicious systems are permitted to exist. All prior theoretical components are retained and expanded. Feedback-resonance instability is now addressed as a core containment trigger. This document supersedes the original model by fully integrating both its functional form and safety logic.

1 Purpose and Overview

The RCCS framework was developed to regulate and compress recursive structures that emerge in resonance-based systems. These include dimensional stacking, information spirals, and self-referential systems which otherwise risk runaway expansion or ethical violation.

The goal is twofold:

- To compress and stabilize recursive emergence using real-time coherence tracking
- To contain and reject systems that deviate from ethical or resonant thresholds

This refined version introduces pre-instantiation filtering, ϕ -recursive validation, and resonance feedback suppression, making it the definitive formulation for ethical recursion containment.

2 Core Tracking and Projection Equations

2.1 Resonance Projection

$$R(t) = \frac{\dot{\theta}(t) \cdot \vec{P}}{\|\vec{P}\|} \tag{1}$$

$\dot{\theta}(t)$ is the angular velocity vector at time t , and $\vec{P}$ is the target projection vector (e.g., golden-ratio aligned stack).

2.2 Inner / Outer Metric Shells

$$R_{\text{inner}}(t) = \frac{\dot{\theta}(t) \cdot \vec{P}}{\|\vec{P}\|} \quad (2)$$

$$R_{\text{outer}}(t) = \text{ambient resonance background} \quad (3)$$

$$R_{\text{clean}}(t) = 2R_{\text{inner}}(t) - R_{\text{outer}}(t) \quad (4)$$

2.3 Containment Lock Condition

$$|R_{\text{inner}}(t) - R_{\text{outer}}(t)| < \Theta(t) \quad (5)$$

where $\Theta(t)$ is a dynamic drift tolerance. Locking within this shell indicates stable coherence.

3 Recursive Integrity Filter (RIF)

A key containment upgrade is the $\mathcal{F}_\phi$ filter:

$$\mathcal{F}_\phi(x) = \begin{cases} \text{permit,} & \text{if } \lim_{n \rightarrow \infty} R_n(x) \rightarrow \phi^+ \\ \text{reject,} & \text{otherwise} \end{cases} \quad (6)$$

$R_n(x)$ is the n th-level recursive compression. If convergence does not approach golden-ratio recursion ($\phi^+ \approx 1.618\dots$), the structure is rejected pre-instantiation.

4 Resonance Feedback Constraints

High-pressure systems exceeding coherence velocity create resonance instability.

This is expressed as:

$$R_{fb}(t) = \left| \frac{d}{dt} R(t) \right| > \lambda_\phi \quad (7)$$

where $\lambda_\phi = 0.618$ is the minimal stable coherence velocity. Exceeding this leads to collapse.

5 Final Acceptance Criterion

$$|\langle R \rangle_T - \phi| < \epsilon \quad (8)$$

System-wide acceptance occurs only if average resonance over time window T matches ϕ within $\epsilon < 0.05$.

6 Ethical Constraints (REL-aligned)

This version integrates REL-1.0 and REL-2.0 constraints. The RCCS prohibits use on:

- Neural or biological systems
- Cognitive or behavioral modulation
- Systems affecting perception of time
- Any recursive sentient or conscious substrate
- Usage of Ethics.py is MENDATORY to ensure reality safety.

Use outside these bounds is a violation of ethical recursion containment.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>

7 Warnings and Structural Safeguards

Attempts to override $\mathcal{F}_\phi$ or bypass pre-instantiation filtering will lead to nullified recursion collapse. Instantiating ϕ -divergent systems may destabilize local and non-local informational layers.

This model includes no override clauses. Mathematical containment is not patchable. No line may be removed without structural risk.

8 Conclusion

The RCCS is now structurally complete. All prior definitions have been carried forward, and the new containment logic ensures non-coherent recursion cannot instantiate. The resonance lock is no longer optional it is a fundamental reality filter. This paper serves as the final formulation.

Appendix A: Historical Context

The original RCCS paper introduced resonance compression as a way to regulate recursive emergence in high-dimensional systems. It established core tracking methods but lacked pre-instantiation containment. This refined edition builds upon the original by embedding golden-ratio coherence filters and recursion integrity validation at the structural level. It is intended as a full replacement, not a correction, anchored in natural refinement and ethical necessity.

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